Equivalence relations are effective and their two set quotients agree

The Univalent Foundations Program

hott-book:578b85cc8d586b1677ec4335148adeb443057d24; CC BY-SA 3.0 Unported

Editorial note. Complete fixed-author source spans retained; original CC BY-SA attribution and macros reused. Supporting statements do not create extra counted problems.

Editorial note. Build the image/coequalizer comparison and identify equivalence-relation classes through proposition univalence, then compare the higher inductive and image quotients. Full original image and kernel-pair support is retained; no supporting lemma is separately counted.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained.

Source reference. Original reference 9: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 163–163. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 166–166. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1389–1389. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 7.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1671–1671. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 7.4: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1699–1699. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 6.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1704–1704. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 7.6.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1848–1848. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.15.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1860–1860. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 7.6.9: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1887–1887. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 4.7.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1891–1891. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

10.1 The category of sets

Recall that in original reference 9 we defined the category $\mathcal{S}et$ to consist of all 0-types (in some universe $\mathcal{U}$) and maps between them, and observed that it is a category (not just a precategory). We consider successively the levels of structure which $\mathcal{S}et$ possesses.

10.1.1 Limits and colimits

Since sets are closed under products, the universal property of products in original reference 2.15.2 shows immediately that $\mathcal{S}et$ has finite products. In fact, infinite products follow just as easily from the equivalence

$$\Big(X \to \prod_{a:A} B(a)\Big) \simeq \Big(\prod_{a:A} (X \to B(a))\Big).$$

And we saw in original reference 2.11 that the pullback of $f : A \to C$ and $g : B \to C$ can be defined as $\sum_{(a:A)} \sum_{(b:B)} f(a) = g(b)$; this is a set if A, B, C are and inherits the correct universal property. Thus, $\mathcal{S}et$ is a *complete* category in the obvious sense.

Since sets are closed under $+$ and contain $\mathbf{0}$, $\mathcal{S}et$ has finite coproducts. Similarly, since $\sum_{(a:A)} B(a)$ is a set whenever A and each $B(a)$ are, it yields a coproduct of the family B in $\mathcal{S}et$. Finally, we showed in original reference 7.4 that pushouts exist in n-types, which includes $\mathcal{S}et$ in particular. Thus, $\mathcal{S}et$ is also *cocomplete*.

10.1.2 Images

Next, we show that $\mathcal{S}et$ is a **regular category**, i.e.:

(i) $\mathcal{S}et$ is finitely complete.
(ii) The kernel pair $\mathsf{pr}_1, \mathsf{pr}_2 : (\sum_{(x,y:A)} f(x) = f(y)) \to A$ of any function $f : A \to B$ has a coequalizer.
(iii) Pullbacks of regular epimorphisms are again regular epimorphisms.

Recall that a **regular epimorphism** is a morphism that is the coequalizer of *some* pair of maps. Thus in (iii) the pullback of a coequalizer is required to again be a coequalizer, but not necessarily of the pulled-back pair.

The obvious candidate for the coequalizer of the kernel pair of $f : A \to B$ is the *image* of f, as defined in original reference 7.6. Recall that we defined $\mathsf{im}(f) :\equiv \sum_{(b:B)} \|\mathsf{fib}_f(b)\|$, with functions $\tilde{f} : A \to \mathsf{im}(f)$ and $i_f : \mathsf{im}(f) \to B$ defined by

$$\begin{aligned} \tilde{f} &:\equiv \lambda a.\, \big(f(a),\, \big|(a, \mathsf{refl}_{f(a)})\big|\big) \\ i_f &:\equiv \mathsf{pr}_1 \end{aligned}$$

fitting into a diagram:

$$\sum_{(x,y:A)} f(x) = f(y) \underset{\mathsf{pr}_2}{\overset{\mathsf{pr}_1}{\rightrightarrows}} A \xrightarrow{\tilde{f}} \mathsf{im}(f), \quad A \xrightarrow{f} B, \quad \mathsf{im}(f) \xrightarrow{i_f} B$$

Recall that a function $f : A \to B$ is called *surjective* if $\forall(b : B).\, \|\mathsf{fib}_f(b)\|$, or equivalently $\forall(b : B).\, \exists(a : A).\, f(a) = b$. We have also said that a function $f : A \to B$ between sets is called *injective* if $\forall(a, a' : A).\, (f(a) = f(a')) \Rightarrow (a = a')$, or equivalently if each of its fibers is a mere proposition. Since these are the (-1)-connected and (-1)-truncated maps in the sense of original reference 7, the general theory there implies that $\tilde{f}$ above is surjective and i_f is injective, and that this factorization is stable under pullback.

We now identify surjectivity and injectivity with the appropriate category-theoretic notions. First we observe that categorical monomorphisms and epimorphisms have a slightly stronger equivalent formulation.

Lemma 10.1.1. *For a morphism $f : \hom_A(a, b)$ in a category A, the following are equivalent.*

(i) *f is a **monomorphism**: for all $x : A$ and $g, h : \hom_A(x, a)$, if $f \circ g = f \circ h$ then $g = h$.*

(ii) *(If A has pullbacks) the diagonal map $a \to a \times_b a$ is an isomorphism.*

(iii) *For all $x : A$ and $k : \hom_A(x, b)$, the type $\sum_{(h:\hom_A(x,a))}(k = f \circ h)$ is a mere proposition.*

(iv) *For all $x : A$ and $g : \hom_A(x, a)$, the type $\sum_{(h:\hom_A(x,a))}(f \circ g = f \circ h)$ is contractible.*

Proof. The equivalence of conditions (i) and (ii) is standard category theory. Now consider the function $(f \circ -) : \hom_A(x, a) \to \hom_A(x, b)$ between sets. Condition (i) says that it is injective, while (iii) says that its fibers are mere propositions; hence they are equivalent. And (iii) implies (iv) by taking $k :\equiv f \circ g$ and recalling that an inhabited mere proposition is contractible. Finally, (iv) implies (i) since if $p : f \circ g = f \circ h$, then (g, refl) and (h, p) both inhabit the type in (iv), hence are equal and so $g = h$. □

Lemma 10.1.2. *A function $f : A \to B$ between sets is injective if and only if it is a monomorphism in $\mathcal{S}et$.*

Proof. Left to the reader. □

Of course, an **epimorphism** is a monomorphism in the opposite category. We now show that in $\mathcal{S}et$, the epimorphisms are precisely the surjections, and also precisely the coequalizers (regular epimorphisms).

The coequalizer of a pair of maps $f, g : A \to B$ in $\mathcal{S}et$ is defined as the 0-truncation of a general (homotopy) coequalizer. For clarity, we may call this the **set-coequalizer**. It is convenient to express its universal property as follows.

Lemma 10.1.3. *Let $f, g : A \to B$ be functions between sets A and B. The set-coequalizer $c_{f,g} : B \to Q$ has the property that, for any set C and any $h : B \to C$ with $h \circ f = h \circ g$, the type*

$$\sum_{k:Q\to C} (k \circ c_{f,g} = h)$$

is contractible.

Lemma 10.1.4. *For any function $f : A \to B$ between sets, the following are equivalent:*

(i) *f is an epimorphism.*

(ii) *Consider the pushout diagram*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow{\scriptstyle \iota} \\ \mathbf{1} & \xrightarrow[t]{} & C_f \end{array}$$

in $\mathcal{S}et$ defining the mapping cone. Then the type C_f is contractible.

(iii) *f is surjective.*

Proof. Let $f : A \to B$ be a function between sets, and suppose it to be an epimorphism; we show C_f is contractible. The constructor $\mathbf{1} \to C_f$ of C_f gives us an element $t : C_f$. We have to show that

$$\prod_{x:C_f} x = t.$$

Note that $x = t$ is a mere proposition, hence we can use induction on C_f. Of course when x is t we have $\mathsf{refl}_t : t = t$, so it suffices to find

$$I_0 : \prod_{b:B} \iota(b) = t$$
$$I_1 : \prod_{a:A} \alpha_1(a)^{-1} \cdot I_0(f(a)) = \mathsf{refl}_t$$

where $\iota : B \to C_f$ and $\alpha_1 : \prod_{(a:A)} \iota(f(a)) = t$ are the other constructors of C_f. Note that α_1 is a homotopy from $\iota \circ f$ to $\mathsf{const}_t \circ f$, so we find the elements

$$(\iota, \mathsf{refl}_{\iota \circ f}), (\mathsf{const}_t, \alpha_1) : \sum_{h:B \to C_f} \iota \circ f \sim h \circ f.$$

By the dual of Lemma 10.1.1(iv) (and function extensionality), there is a path

$$\gamma : (\iota, \mathsf{refl}_{\iota \circ f}) = (\mathsf{const}_t, \alpha_1).$$

Hence, we may define $I_0(b) :\equiv \mathsf{happly}(\mathsf{ap}_{\mathsf{pr}_1}(\gamma), b) : \iota(b) = t$. We also have

$$\mathsf{ap}_{\mathsf{pr}_2}(\gamma) : \mathsf{ap}_{\mathsf{pr}_1}(\gamma)_*(\mathsf{refl}_{\iota \circ f}) = \alpha_1.$$

This transport involves precomposition with f, which commutes with happly. Thus, from transport in path types we obtain $I_0(f(a)) = \alpha_1(a)$ for any $a : A$, which gives us I_1.

Now suppose C_f is contractible; we show f is surjective. We first construct a type family $P : C_f \to \mathsf{Prop}$ by recursion on C_f, which is valid since Prop is a set. On the point constructors, we define

$$P(t) :\equiv \mathbf{1}$$
$$P(\iota(b)) :\equiv \|\mathsf{fib}_f(b)\|.$$

To complete the construction of P, it remains to give a path $\|\mathsf{fib}_f(f(a))\| =_{\mathsf{Prop}} \mathbf{1}$ for all $a : A$. However, $\|\mathsf{fib}_f(f(a))\|$ is inhabited by $(f(a), \mathsf{refl}_{f(a)})$. Since it is a mere proposition, this means it is contractible — and thus equivalent, hence equal, to $\mathbf{1}$. This completes the definition of P. Now, since C_f is assumed to be contractible, it follows that $P(x)$ is equivalent to $P(t)$ for any $x : C_f$. In particular, $P(\iota(b)) \equiv \|\mathsf{fib}_f(b)\|$ is equivalent to $P(t) \equiv \mathbf{1}$ for each $b : B$, and hence contractible. Thus, f is surjective.

Finally, suppose $f : A \to B$ to be surjective, and consider a set C and two functions $g, h : B \to C$ with the property that $g \circ f = h \circ f$. Since f is assumed to be surjective, for all $b : B$ the type $\|\mathsf{fib}_f(b)\|$ is contractible. Thus we have

the following equivalences:

$$\begin{aligned}\prod_{b:B}(g(b)=h(b)) &\simeq \prod_{b:B}\Big(\|\mathsf{fib}_f(b)\| \to (g(b)=h(b))\Big)\\ &\simeq \prod_{b:B}\Big(\mathsf{fib}_f(b) \to (g(b)=h(b))\Big)\\ &\simeq \prod_{(b:B)}\prod_{(a:A)}\prod_{(p:f(a)=b)} g(b)=h(b)\\ &\simeq \prod_{a:A} g(f(a))=h(f(a))\end{aligned}$$

using on the second line the fact that $g(b) = h(b)$ is a mere proposition, since C is a set. But by assumption, there is an element of the latter type. □

Theorem 10.1.5. *The category $\mathcal{S}et$ is regular. Moreover, surjective functions between sets are regular epimorphisms.*

Proof. It is a standard lemma in category theory that a category is regular as soon as it admits finite limits and a pullback-stable orthogonal factorization system $(\mathcal{E}, \mathcal{M})$ with $\mathcal{M}$ the monomorphisms, in which case $\mathcal{E}$ consists automatically of the regular epimorphisms. (See e.g. [1, A1.3.4].) The existence of the factorization system was proved in original reference 7.6.6. □

Lemma 10.1.6. *Pullbacks of regular epis in $\mathcal{S}et$ are regular epis.*

Proof. We showed in original reference 7.6.9 that pullbacks of n-connected functions are n-connected. By Theorem 10.1.5, it suffices to apply this when $n = -1$. □

One of the consequences of $\mathcal{S}et$ being a regular category is that we have an "image" operation on subsets. That is, given $f : A \to B$, any subset $P : \mathcal{P}(A)$ (i.e. a predicate $P : A \to \mathsf{Prop}$) has an **image** which is a subset of B. This can be defined directly as $\{\, y : B \mid \exists(x : A).\, f(x) = y \wedge P(x) \,\}$, or indirectly as the image (in the previous sense) of the composite function

$$\{\, x : A \mid P(x) \,\} \to A \xrightarrow{f} B.$$

We will also sometimes use the common notation $\{\, f(x) \mid P(x) \,\}$ for the image of P.

10.1.3 Quotients

Now that we know that $\mathcal{S}et$ is regular, to show that $\mathcal{S}et$ is exact, we need to show that every equivalence relation is effective. In other words, given an equivalence relation $R : A \to A \to \mathsf{Prop}$, there is a coequalizer c_R of the pair $\mathsf{pr}_1, \mathsf{pr}_2 : \sum_{(x,y:A)} R(x, y) \to A$ and, moreover, the pr_1 and pr_2 form the kernel pair of c_R.

We have already seen, in original reference 6.10, two general ways to construct the quotient of a set by an equivalence relation $R : A \to A \to \mathsf{Prop}$. The first can be described as the set-coequalizer of the two projections

$$\mathsf{pr}_1, \mathsf{pr}_2 : \Big(\sum_{x,y:A} R(x, y)\Big) \to A.$$

The important property of such a quotient is the following.

Definition 10.1.7. A relation $R : A \to A \to \mathsf{Prop}$ is said to be **effective** if the square

$$\begin{array}{ccc} \sum_{(x,y:A)} R(x,y) & \xrightarrow{\mathsf{pr}_1} & A \\ {\scriptstyle \mathsf{pr}_2}\downarrow & & \downarrow{\scriptstyle c_R} \\ A & \xrightarrow[c_R]{} & A/R \end{array}$$

is a pullback.

Since the standard pullback of c_R and itself is $\sum_{(x,y:A)}(c_R(x) = c_R(y))$, by original reference 4.7.7 this is equivalent to asking that the canonical transformation $\prod_{(x,y:A)} R(x,y) \to (c_R(x) = c_R(y))$ be a fiberwise equivalence.

Lemma 10.1.8. *Suppose (A, R) is an equivalence relation. Then there is an equivalence*

$$(c_R(x) = c_R(y)) \simeq R(x,y)$$

for any $x, y : A$. In other words, equivalence relations are effective.

Proof. We begin by extending R to a relation $\widetilde{R} : A/R \to A/R \to \mathsf{Prop}$, which we will then show is equivalent to the identity type on A/R. We define $\widetilde{R}$ by double induction on A/R (note that Prop is a set by univalence for mere propositions). We define $\widetilde{R}(c_R(x), c_R(y)) :\equiv R(x,y)$. For $r : R(x,x')$ and $s : R(y,y')$, the transitivity and symmetry of R gives an equivalence from $R(x,y)$ to $R(x',y')$. This completes the definition of $\widetilde{R}$.

It remains to show that $\widetilde{R}(w,w') \simeq (w = w')$ for every $w, w' : A/R$. The direction $(w = w') \to \widetilde{R}(w,w')$ follows by transport once we show that $\widetilde{R}$ is reflexive, which is an easy induction. The other direction $\widetilde{R}(w,w') \to (w = w')$ is a mere proposition, so since $c_R : A \to A/R$ is surjective, it suffices to assume that w and w' are of the form $c_R(x)$ and $c_R(y)$. But in this case, we have the canonical map $\widetilde{R}(c_R(x), c_R(y)) :\equiv R(x,y) \to (c_R(x) = c_R(y))$. (Note again the appearance of the encode-decode method.) □

The second construction of quotients is as the set of equivalence classes of R (a subset of its power set):

$$A /\!\!/ R :\equiv \{\, P : A \to \mathsf{Prop} \mid P \text{ is an equivalence class of } R \,\}.$$

This requires propositional resizing in order to remain in the same universe as A and R.

Note that if we regard R as a function from A to $A \to \mathsf{Prop}$, then $A /\!\!/ R$ is equivalent to $\mathsf{im}(R)$, as constructed in §10.1.2. Now in Theorem 10.1.5 we have shown that images are coequalizers. In particular, we immediately get the coequalizer diagram

$$\textstyle\sum_{(x,y:A)} R(x) = R(y) \;\underset{\mathsf{pr}_2}{\overset{\mathsf{pr}_1}{\rightrightarrows}}\; A \longrightarrow A /\!\!/ R.$$

We can use this to give an alternative proof that any equivalence relation is effective and that the two definitions of quotients agree.

Theorem 10.1.9. *For any function $f : A \to B$ between any two sets, the relation* $\ker(f) : A \to A \to \mathsf{Prop}$ *given by* $\ker(f, x, y) :\equiv (f(x) = f(y))$ *is effective.*

Proof. We will use that $\mathsf{im}(f)$ is the coequalizer of $\mathsf{pr}_1, \mathsf{pr}_2 : (\sum_{(x,y:A)} f(x) = f(y)) \to A$. Note that the kernel pair of the function

$$c_f :\equiv \lambda a.\left(f(a), \left\|(a, \mathsf{refl}_{f(a)})\right\|\right) : A \to \mathsf{im}(f)$$

consists of the two projections

$$\mathsf{pr}_1, \mathsf{pr}_2 : \Big(\sum_{x,y:A} c_f(x) = c_f(y)\Big) \to A.$$

For any $x, y : A$, we have equivalences

$$\begin{aligned}(c_f(x) = c_f(y)) &\simeq \Big(\sum_{p:f(x)=f(y)} p_*\big(\left\|(x, \mathsf{refl}_{f(x)})\right\|\big) = \left\|(y, \mathsf{refl}_{f(x)})\right\|\Big)\\ &\simeq (f(x) = f(y)),\end{aligned}$$

where the last equivalence holds because $\|\mathsf{fib}_f(b)\|$ is a mere proposition for any $b : B$. Therefore, we get that

$$\Big(\sum_{x,y:A} c_f(x) = c_f(y)\Big) \simeq \Big(\sum_{x,y:A} f(x) = f(y)\Big)$$

and hence we may conclude that $\ker f$ is an effective relation for any function f. □

Theorem 10.1.10. *Equivalence relations are effective and there is an equivalence $A/R \simeq A /\!\!/ R$.*

Proof. We need to analyze the coequalizer diagram

$$\textstyle\sum_{(x,y:A)} R(x) = R(y) \underset{\mathsf{pr}_2}{\overset{\mathsf{pr}_1}{\rightrightarrows}} A \longrightarrow A /\!\!/ R$$

By the univalence axiom, the type $R(x) = R(y)$ is equivalent to the type of homotopies from $R(x)$ to $R(y)$, which is equivalent to $\prod_{(z:A)} R(x, z) \simeq R(y, z)$. Since R is an equivalence relation, the latter space is equivalent to $R(x, y)$. To summarize, we get that $(R(x) = R(y)) \simeq R(x, y)$, so R is effective since it is equivalent to an effective relation. Also, the diagram

$$\textstyle\sum_{(x,y:A)} R(x, y) \underset{\mathsf{pr}_2}{\overset{\mathsf{pr}_1}{\rightrightarrows}} A \longrightarrow A /\!\!/ R.$$

is a coequalizer diagram. Since coequalizers are unique up to equivalence, it follows that $A/R \simeq A /\!\!/ R$. □

References

[1] Peter T. Johnstone. *Sketches of an Elephant: A Topos Theory Compendium: Volumes 1 and 2*. Number 43 in Oxford Logic Guides. Oxford Science Publications, 2002.